

46 YEARS OF EXCELLENCE

NARAYANA
IIT - JEE ACADEMY - INDIA

Sec: SR.IIT _*CO-SC(MODEL-A)
Time: 3 Hrs

Date: 23-11-25
Max. Marks: 300

23-11-25_SR.IIT_STAR CO-SC(MODEL-A)_JEE MAIN_GTM-3_SYLLABUS

MATHEMATICS: TOTAL SYLLABUS

PHYSICS: TOTAL SYLLABUS

CHEMISTRY: TOTAL SYLLABUS

Rankers Academy JEE
THE NARAYANA GROUP
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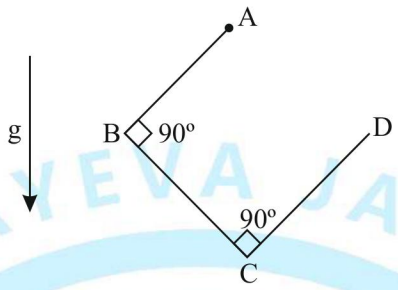
PHYSICS **MAX.MARKS: 100**

SECTION – I
(SINGLE CORRECT ANSWER TYPE)

This section contains 20 multiple choice questions. Each question has 4 options (1), (2), (3) and (4) for its answer, out of which **ONLY ONE** option can be correct.

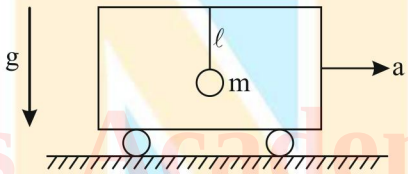
Marking scheme: +4 for correct answer, 0 if not attempted and -1 if not correct.

1. Three identical rods are hinged at point A in vertical plane as shown. The angle made by rod AB with vertical is



A) $\tan^{-1}\left(\frac{1}{\sqrt{3}}\right)$ B) $\tan^{-1}\left(\frac{3}{4}\right)$ C) $\tan^{-1}(1)$ D) $\tan^{-1}\left(\frac{4}{3}\right)$

2. In shown figure, the trolley starts accelerating with acceleration a. The maximum angle deflected by thread from vertical will be.

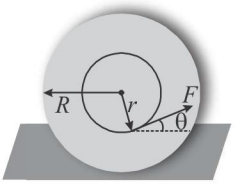


A) $\tan^{-1}\left(\frac{a}{g}\right)$ B) $\tan^{-1}\left(\frac{2a}{g}\right)$ C) $2\tan^{-1}\left(\frac{a}{g}\right)$ D) $\tan^{-1}\left(\frac{a}{2g}\right)$

3. A particle moves in xy-plane. The position vector of particle at any time t is $\vec{r}=\left\{(2t)\hat{i}+\left(2t^2\right)\hat{j}\right\}m$. The rate of change of θ at time $t=2$ second. (where θ is the angle which its velocity vector makes with positive x-axis) is:

A) $\frac{2}{17}$ rad / s B) $\frac{1}{14}$ rad / s C) $\frac{4}{7}$ rad / s D) $\frac{6}{5}$ rad / s

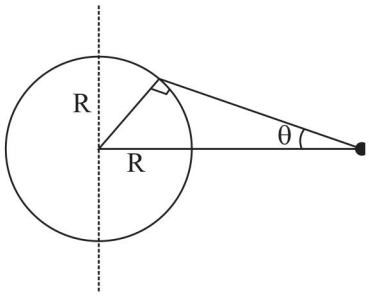
4. The spool shown in figure placed on a rough horizontal surface has inner radius r and outer radius R. A tangential force is applied on inner surface having radius r. The angle θ between the applied force and the horizontal can be varied. The critical angle (θ) for which the spool does not roll and remains stationary is given by



A) $\sin\theta=\frac{r}{R}$ B) $\cos\theta=\frac{r}{R}$ C) $\cos\theta=\sqrt{\frac{r}{R}}$ D) $\cos\theta=\frac{2r}{R}$

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5. A geostationary satellite S is at a height h above the surface of earth. If earth radius is R



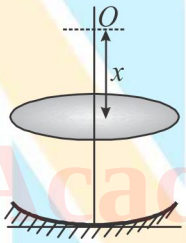
A) The area on earth escaped from this satellite is given as $2\pi R^2(1-\sin\theta)$

B) The area on earth escaped from this satellite is given as $2\pi R^2(1-\cos\theta)$

C) The area on earth escaped from this satellite is given as $2\pi R^2(1+\sin\theta)$

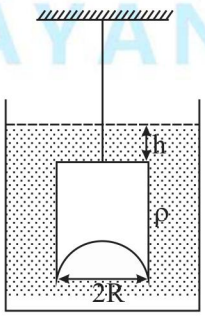
D) The area on earth escaped from this satellite is given as $2\pi R^2(1+\cos\theta)$

6. A thin convex lens of focal length 40 cm is held co-axially 12 cm above a mirror of focal length 18 cm. A real object held x cm above the lens gives rise to an image coincident with it. Then x is equal to:



A) 12 cm B) 15 cm C) 18 cm D) 30 cm

7. A hemispherical portion of radius R is removed from the bottom of a cylinder of radius R. The volume of the remaining cylinder is V and its mass M. It is suspended by a string in a liquid of density ρ where it stays vertical. The upper surface of the cylinder is at a depth h below the liquid surface. The force on the bottom of the cylinder by the liquid is



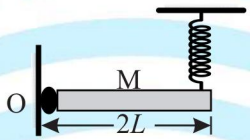
A) Mg B) $Mg-V\rho g$ C) $Mg+\pi R^2h\rho g$ D) $\rho g\left(V+\pi R^2h\right)$

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8. A Stationary body of mass m_1+m_2 is split into two parts of masses m_1 and m_2 by an internal explosion which generates a kinetic energy E. If after explosion the parts move, then their relative speed is

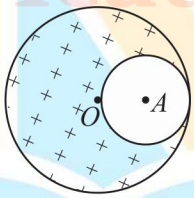
A) $\sqrt{\frac{E(m_1+m_2)}{m_1+m_2}}$ B) $\sqrt{\frac{2E(m_1+m_2)}{m_1m_2}}$ C) $\sqrt{\frac{E(m_1+m_2)}{m_1+m_2}}$ D) $\sqrt{\frac{E(m_1+m_2)}{2m_1\times m_2}}$

9. A uniform pole of length $2L$ is laid on a smooth horizontal table as shown in figure. The mass of pole is M and it is connected to a frictionless axis at O. A spring with force constant k is connected to the other end. The pole is displaced by a small angle θ from equilibrium position and released such that it performs small oscillations in horizontal plane. Its angular frequency of oscillations is



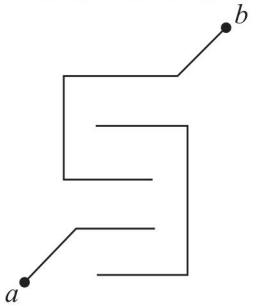
A) $\omega=\sqrt{\frac{M}{3k}}$ B) $\omega=\sqrt{\frac{k}{3M}}$ C) $\omega=\sqrt{\frac{3k}{M}}$ D) $\omega=\sqrt{\frac{k}{2M}}$

10. A solid sphere having uniform charge density ρ and radius R is shown in figure. A spherical cavity of radius $\frac{R}{2}$ is hollowed out. What is potential at point O ? (Assuming potential of infinity to be zero)



A) $\frac{11R^2\rho}{24\varepsilon_0}$ B) $\frac{5}{12}\frac{R^2\rho}{\varepsilon_0}$ C) $\frac{7\rho R^2}{12\varepsilon_0}$ D) $\frac{3}{2}\frac{R^2\rho}{\varepsilon_0}$

11. Plates of area A are arranged as shown. The distance between each plate is d, the net capacitance is (d << A)



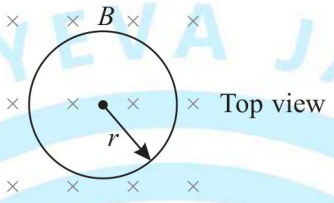
A) $\frac{\varepsilon_0 A}{d}$ B) $\frac{7\varepsilon_0 A}{d}$ C) $\frac{6\varepsilon_0 A}{d}$ D) $\frac{5\varepsilon_0 A}{3d}$

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12. In a nuclear reactor an element X decays to a radioactive element Y at a constant rate 10^{15} atoms per sec. Each decay releases 100 MeV energy. Y immediately decays to a stable product Z. Each decay of Y releases 50 MeV. All energy released inside the reactor is used to produced electricity at an efficiency of 25%. Calculate the electrical power in kW generated in the reactor in steady state. (Take $\text{eV} = 1.6\times10^{-19}\ \text{J}$)

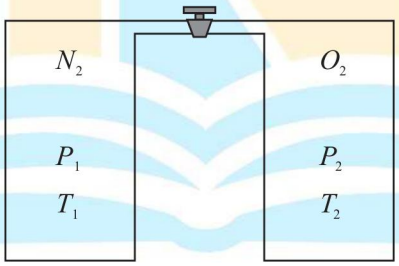
A) 6 B) 7 C) 8 D) 5

13. A nonconducting ring (of mass m, radius r, having charge Q) is placed on a rough horizontal surface (in a cylindrical region with transverse magnetic field). The field is increasing with time at the rate R and coefficient of friction between the surface and the ring is μ. For ring to remain in equilibrium minimum value of μ should be



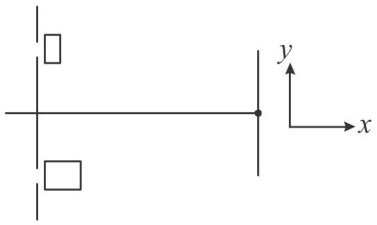
A) $\frac{Qr^2R^2}{2mg}$ B) $\frac{QrR}{2mg}$ C) $\frac{Qr^2R}{2mg}$ D) $\frac{QrR^2}{2mg}$

14. Two identical vessels are connected by a tube. Volume of the tube is vanishingly small and it is equipped with a valve. One of the vessels contains nitrogen at pressure P_1 and temperature T_1 and the other vessel contains oxygen at pressure P_2 and temperature T_2. The system is well insulated from the surroundings. Now the valve is opened. The final temperature T is attained by mixture of both the gases. Then (Assuming gases to be ideal and non-interactive)



A) $T=\frac{P_1T_1+P_2T_2}{P_1+P_2}$ B) $T=\frac{P_1T_2+P_2T_1}{P_1+P_2.}$ B) $T=\frac{(P_1+P_2)T_1T_2}{2(P_1T_2+P_2T_1)}$ D) $T=\frac{(P_1+P_2)T_1T_2}{P_1T_2+P_2T_1}$

15. In YDSE, both slits are covered by transparent slab. Upper slit is covered by slab of refractive index 1.5 and thickness t and lower is covered by refractive index $\frac{4}{3}$ and thickness 2t, then central maxima



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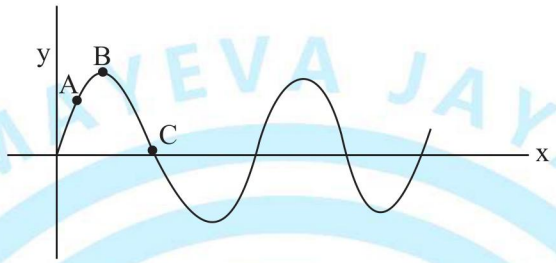
A) shifts in +ve y-axis direction

B) shifts in –ve y-axis direction

C) remains at same position

D) may shift in upward or downward depending upon wavelength of light

16. A plane progressive transverse wave is passing through medium along the positive x-axis at the given instant of time. The nature of the displacement - position (y vs x) graph of particles of medium at that instant is as shown in figure. Which of the following statements are correct. (A, B, C are three particles of the vibrating string)



(1) Elastic potential energy per unit length of the string at the point B is more than that at points A and C

(2) Elastic potential energy per unit length of the string at the point C is more than that at points A and B

(3) velocity of the particle at A is along negative y -axis.

(4) Velocity of the particle at C is along positive y -axis.

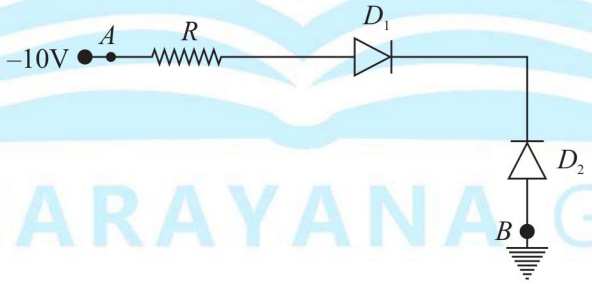
A) Only 2,3,4 are correct

B) Only 1,3,4 are correct

C) Only 1,3 are correct

D) Only 1,4 are correct

17. In figure, assuming the diodes to be ideal, then choose correct option



A) D_1 is forward biased and D_2 is reverse biased and hence current flows from A to B.

B) D_2 is forward biased and D_1 is reverse biased and hence no current flows from B to A and vice-versa.

C) D_1 and D_2 are both forward biased and hence current flows from A to B.

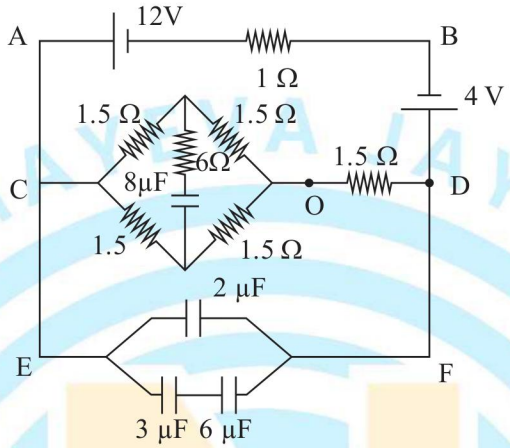
D) D_1 and D_2 are both reverse biased and hence no current flows from A to B and vice-versa.

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18. A solid copper cube of edges 1 cm is suspended in an evacuated enclosure. Its temperature is found to fall from $100^{\circ}C$ to $99^{\circ}C$ in 100 s . Another solid copper cube of edges 2 cm, with similar surface nature, is suspended in a similar manner. The time required for this cube to cool from $100^{\circ}C$ to $99^{\circ}C$ will be approximately

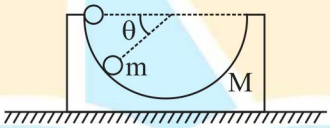
A) 100 sec B) 130 sec C) 150 sec D) 200 sec

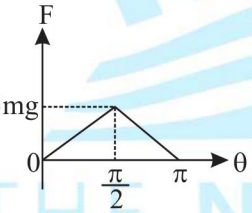
19. In the given circuit, the potential difference (in V) across the 6μF capacitor in steady state is

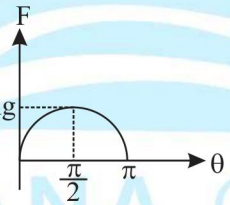


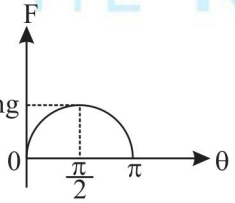
A) 2 V B) 4 V C) 6 V D) 8 V

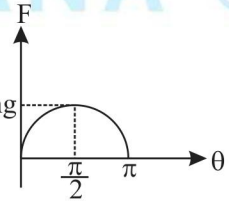
20. In the shown figure, a small mass m starts sliding down a smooth stationary circular track. Which of the following graph best represents the variation of magnitude of the force applied by the track on the mass and the angle θ ?



A) 

B) 

C) 

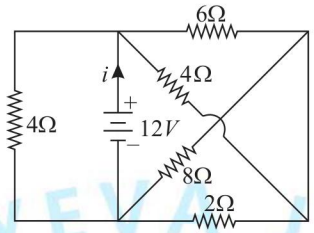
D) 

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SECTION-II
(NUMERICAL VALUE ANSWER TYPE)

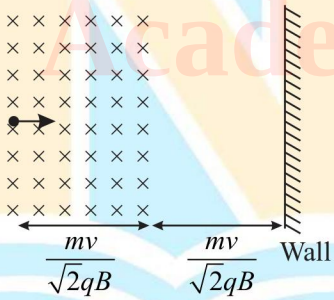
This section contains 5 questions. The answer to each question is a Numerical value. If the Answer in the decimals, **Round off to the nearest Integer only. Have to Answer all 5 questions** and question will be evaluated according to the following marking scheme:
Marking Scheme: +4 for correct answer, 0 if not attempted and –1 if not correct.

21. Compute the value of battery current (in A) in the circuit shown in figure. All the resistances are in ohm.

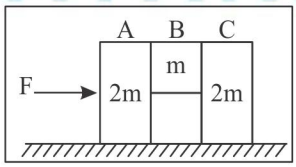


22. A glass capillary tube of inner diameter 0.28 mm is lowered vertically into water in a vessel. The total pressure to be applied on the water in the tube so that water level in the tube is same as that in the vessel is $n\times10^3\ N/m^2$. (surface tension of water $=0.07\ N/m$, atmospheric pressure $=10^5\ N/m^2$ Angle of contact is 0°) : Find value of n.

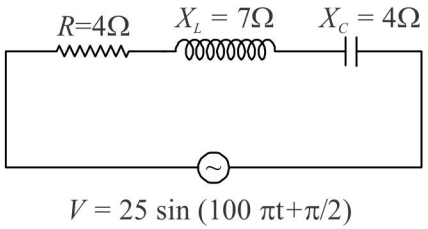
23. A particle of mass m and charge q enters a region of magnetic field (as shown) with speed v. There is a region in which the magnetic field is absent, as shown. The particle after entering the region collides elastically with a rigid wall. Time after which the velocity of particle becomes antiparallel to its initial velocity is $\frac{m}{2qB}(\pi+n)$ find the value of n. (gravity is neglected)



24. The system is pushed by the force F as shown. All surfaces are smooth expect between $B\&C$. Friction coefficient between $B\&C$ is μ. Minimum value of F to prevent block B from downward slipping is: $n\left(\frac{mg}{2\mu}\right)$, find value of n.



25. For the series $L-C-R$ circuit as shown in figure, the heat developed in 80 s is (in J)



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CHEMISTRY **MAX.MARKS: 100**

SECTION – I
(SINGLE CORRECT ANSWER TYPE)

This section contains 20 multiple choice questions. Each question has 4 options (1), (2), (3) and (4) for its answer, out of which **ONLY ONE** option can be correct.

Marking scheme: +4 for correct answer, 0 if not attempted and -1 if not correct.

26. For the following equilibrium, $N_2(g)+3H_2(g) \rightleftharpoons 2NH_3(g)$

Calculate the temperature at which $\frac{K_P(atm^{-2})}{K_C(M^{-2})}=2$?

A) 0.027 K B) 0.36 K C) 36.54 K D) 8.61 K

27. For reaction between reactants A and B the following data is obtained-

S.No.	[A]	[B]	Rate
	$mol.L^{-1}$	$mol.L^{-1}$	$mol.L^{-1}\,sec^{-1}$
I	1×10^{-2}	2×10^{-2}	2×10^{-4}
II	2×10^{-2}	2×10^{-2}	4×10^{-4}
III	2×10^{-2}	4×10^{-2}	8×10^{-4}

Which of the following are correct statements?

(a) Rate constant of the reaction is 10^{-8}

(b) Overall order of the reaction is two

(c) Rate of reaction increases four times on doubling the concentration of both the reactant.

A) a, b and c B) a and b C) b and c D) c alone

28. If standard reduction potentials of A^{2+}/A and B^{3+}/B couples are –1V and 2V respectively, then E_{cell} for $A|A^{2+}(0.01M)||B^{3+}(0.1M)|B$ will be :-

A) $>E^{o}_{cell}$ B) $=E^{o}_{cell}$ C) $<E^{o}_{cell}$ D) unpredictable

29. How many gm of solid $Ca(OH)_2$ must be added to 100 mL of a buffer solution? Which is 0.1 M each w.r.t. acid HA and salt KA to make the pH of solution 6.0. [Given : $pK_a(HA)=5$] (assume volume does not change on adding $Ca(OH)_2$):-

[Atomic mass of Ca = 40 g/mol]

A) 0.5 B) 0.4 C) 0.3 D) 0.2

30. In a compound Carbon = 40%, Hydrogen = 6.66% and rest is oxygen by mass are present. The molecular formula of the compound could be -

A) $C_{10}H_{20}O_{16}$ B) $C_{40}H_{80}O_{32}$ C) $C_6H_{12}O_6$ D) $C_2H_6O_3$

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31. To completely oxidize a mixture of 1 mole of $FeSO_4$, 3 moles of FeC_2O_4 and 4 moles of $Fe_2(C_2O_4)_3$ we require x moles of $K_2Cr_2O_7$ in acidic medium. Calculate x.

A) $\frac{10}{3}$ B) $\frac{17}{3}$ C) $\frac{22}{3}$ D) $\frac{17}{6}$

32. For the reactions, $2A + 3B \rightarrow C$, $\Delta H = 200\,kJ\,mol^{-1}$ and $\Delta S = 2000\,J\,K^{-1}\,mol^{-1}$.

Calculate temperature (in K) below which reaction become non-spontaneous?

(considering ΔH and ΔS to be constant over the temperature range) :-

A) 0.1 B) 110 C) 100 D) 1000

33. Which of the following can rotate plane polarized light?

A) $[Co(en)(NH_3)_2]^{2+}$ B) $[Co(H_2O)_4(en)]^{3+}$

C) $[Co(en)_2(NH_3)_2]^{3+}$ D) $[Co(NH_3)_3Cl]^{+}$

34. In B_2H_6 the maximum number of atoms in a single plane are-

A) 4 B) 6 C) 8 D) 5

35. Which of the following statements is correct?

A) Copper liberates hydrogen from acids.

B) In its higher oxidation states, manganese forms unstable compounds with oxygen and fluorine.

C) Mn^{3+} and Co^{3+} are oxidising agents in aqueous solution.

D) Ti^{2+} and Cr^{2+} are oxidizing agents in aqueous solution.

36. If radius of 5th orbit of Li^{2+} ion is x. Calculate radius of 2nd excited state of Be^{+3} ion-

A) $\frac{9x}{10}$ B) $\frac{27x}{100}$ C) $\frac{8x}{9}$ D) $\frac{9x}{8}$

37. Which of the following set of elements have almost the same radii-

A) Y & La B) Ti & Zr C) Mo & W D) V & Nb

38. Which of the following ion has maximum complex forming tendency?

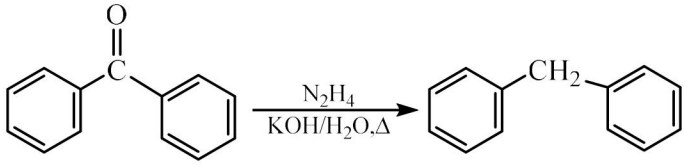
A) La^{+3} B) Nd^{+3} C) Eu^{+3} D) Lu^{+3}

39. The correct dipole moment order is-

A) $HF > H_2O > NH_3 > H_2S$ B) $H_2O > HF > NH_3 > H_2S$

C) $HF > H_2O > H_2S > NH_3$ D) $HF > H_2S > H_2O > NH_3$

For More Join: @IITJEE_Advanced

40. 

This reduction reaction is known as:

A) Rosenmund reduction B) Wolff-Kishner reduction

C) Stephen reduction D) Etard reduction

41. Cellulose has monomer and linkages as-

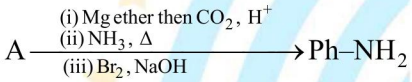
A) Monomer as β-D-glucose & β-1, 4-glycosidic linkage

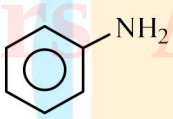
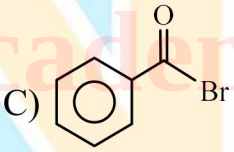
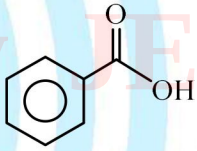
B) Monomer as β-D-glucose & α-1, 4-glycosidic linkage

C) Monomer as α-D-glucose & β-1, 4-glycosidic linkage

D) Monomer as α-D-glucose & α-1, 4-glycosidic linkage

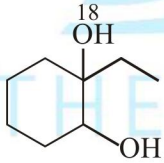
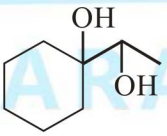
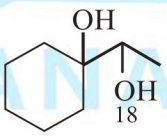
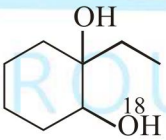
42. Aniline as final product is formed in the following multistep reaction sequence if starting reactant is taken as-



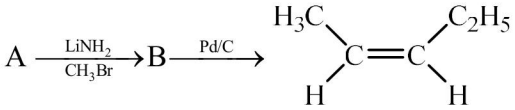
A)  B)  C)  D) 

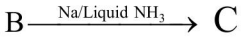
43. 

X is

A)  B)  C)  D) 

44. In the given reactions identify A and C.



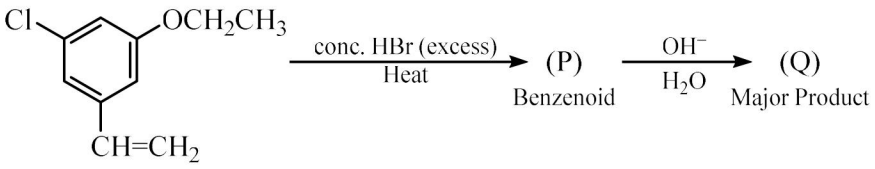


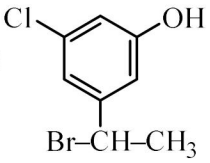
A) A : 2 – Butyne C : Cis - 2 - pentene B) A : Butyne C : trans - 2- pentene

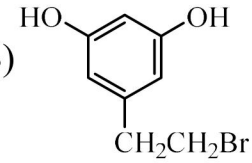
C) A : 2 – Pentyne C : Cis - 2 - butene D) A : Butyne C : Cis - 2 – pentene

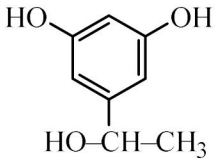
For More Join: @IITJEE_Advanced

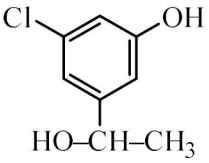
45. The major product (P) in the following reaction is-



A) 

B) 

C) 

D) 

SECTION-II
(NUMERICAL VALUE ANSWER TYPE)

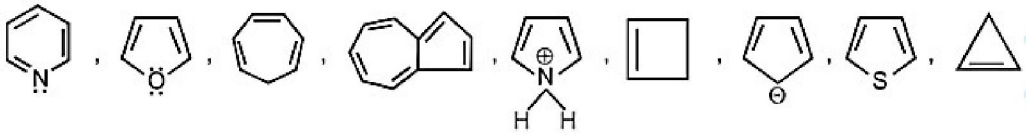
This section contains 5 questions. The answer to each question is a Numerical value. If the Answer in the decimals, **Round off to the nearest Integer only. Have to Answer all 5 questions** and question will be evaluated according to the following marking scheme:
Marking Scheme: +4 for correct answer, 0 if not attempted and –1 if not correct.

46. A solution of A and B has a vapour pressure 200 mm of Hg if it has 25 % mole of A. Another solution of same components containing 25% mole B has vapour pressure 300 mm of Hg. Calculate vapour pressure (in mm Hg) of solution made by mixing 4 moles of A and 1 mole of non-volatile solute C.

47. Calculate the magnitude of enthalpy change for the reaction in Kcal/mol $3H_2(g)+C_6H_6(g)\rightarrow C_6H_{12}(g)$. The bond energies of H_2, $C-H$, $C-C$, $C=C$, and magnitude of resonance energy of benzene are, 110, 100, 80, 150, 36 Kcal/mole respectively.

48. Total number of Monochlorinated products (including stereoisomers) are formed on photo-chlorination of neo-hexane

49. How many species out of the following are aromatic?



50. If reductive ozonolysis of one mole of compound A produces 2 moles of formaldehyde and two moles of glyoxal only. Calculate degree of unsaturation of compound A.

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MATHEMATICS **MAX.MARKS: 100**

SECTION – I
(SINGLE CORRECT ANSWER TYPE)

This section contains 20 multiple choice questions. Each question has 4 options (1), (2), (3) and (4) for its answer, out of which **ONLY ONE** option can be correct.

Marking scheme: +4 for correct answer, 0 if not attempted and -1 if not correct.

51. If $a_1, a_2, \ldots a_{10}$ are positive numbers in arithmetic progression such that $\frac{1}{a_1a_2}+\frac{1}{a_2a_3}+..+\frac{1}{a_9a_{10}}=\frac{9}{64}$ and $\frac{1}{a_1a_{10}}+\frac{1}{a_2a_9}+\ldots\ldots\frac{1}{a_{10}a_1}=\frac{1}{10}\left(\frac{1}{a_1}+\ldots.+\frac{1}{a_{10}}\right)$, then sum of digits of $\left(4\left(\frac{a_1}{a_{10}}+\frac{a_{10}}{a_1}\right)\right)$ is

A) 5 B) 6 C) 7 D) 8

52. If α αnd β are the roots of the equation $x^2+px+q=0$ and also $x^{3900}+p^{1950}x^{1950}+q^{1950}=0$ and if $\frac{\alpha}{\beta},\frac{\beta}{\alpha}$ are the roots of $x^n+1+(x+1)^n=0$, then the value of 'n' must be equal to?

A) 1945 B) 1955 C) 1950 D) 1960

53. Let $a=\text{cosec}10^\circ+\sec 40^\circ-\text{cosec}70^\circ$ and $b=4\cos 18^\circ-3\text{cosec}72^\circ-2\tan 18^\circ$ then $a+2b$ is equal to:

A) $3+\sqrt{5}$ B) $2\sqrt{3}+\sqrt{5}$ C) 7 D) 6

54. If $f(x)=-1+ax+a$ neither touches nor cuts the curve $f(x)=\ell nx$, then minimum value of 'a' may lie in:

A) $\left(\frac{1}{e},\frac{1}{\sqrt{e}}\right)$ B) (e,e^2) C) $\left(\frac{1}{\sqrt{e}},e\right)$ D) (e^2,e^3)

55. Consider two circles $x^2+y^2-4x-4y+4=0$ and $x^2+y^2-16x-4y+59=0$. A line $y=3x+\alpha$ is such that it lies between two circles without touching or intersecting either of them. If 'α' belongs to $(a\sqrt{10}-b,c\sqrt{10}+d)$ then $a+b+c+d$ equals :

A) 23 B) 19 C) 31 D) 27

56. $\int\frac{\text{cosec}^2x-2005}{\cos^{2005}x}dx$ is equal to (where 'C' is an arbitrary constant)

A) $\frac{\cot x}{(\cos x)^{2005}}+C$ B) $\frac{\tan x}{(\cos x)^{2005}}+C$ C) $\frac{-(\tan x)}{(\cos x)^{2005}}+C$ D) $-\frac{\cot x}{(\cos x)^{2005}}+C$

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57. The solution of the equation $\left(x^2y+x^2\right)dx+y^2(x-1)dy=0$ is given by (where 'C' is an arbitrary constant)

A) $x^2+y^2+2(x-y)+2\ln\frac{(x-1)(y+1)}{C}=0$ B) $x^2+y^2+2(x-y)+\ln\frac{(x-1)(y+1)}{C}=0$

C) $x^2+y^2+2(x-y)-2\ln\frac{(x-1)(y+1)}{C}=0$ D) $x^2+y^2+2(x-y)-\ln\frac{(x-1)(y+1)}{C}=0$

58. The area bounded by $y\le 3-|3-x|$ and $y\ge|x-3|$ is

A) $\frac{5}{2}$ B) $\frac{7}{2}$ C) $\frac{9}{2}$ D) $\frac{11}{2}$

59. The value of $\int_{-3}^{-1}(x+1)^4(x+2)^3(x+3)^4dx$ is

A) 1 B) 2 C) 3 D) 0

60. Relation $R=\{(x,y):x$ and y are integers and $x^2+y^2=64\}$ is

A) Reflexive B) Symmetric C) Transitive D) Equivalence

61. If $\vec{a},\vec{b}$ and $\vec{c}$ are vectors such that $\left[\vec{a}\ \ \vec{b}\ \ \vec{c}\right]=4$, then the value of $\left[(2\vec{a}\times3\vec{b})\ \ (3\vec{b}\times4\vec{c})\ \ (4\vec{c}\times2\vec{a})\right]$ is (where $\left[\vec{\alpha}\ \ \vec{\beta}\ \ \vec{\gamma}\right]=(\vec{\alpha}\times\vec{\beta})\cdot\vec{\gamma}=\vec{\alpha}\cdot(\vec{\beta}\times\vec{\gamma})$)

A) $2^{10}\times9$ B) $2^9\times9$ C) $2^8\times9$ D) $2^{11}\times9$

62. If $\tan^{-1}\left(x^2+3|x|-4\right)=\tan^{-1}\left(4\pi+\sin^{-1}(\sin14)\right)$ then the value of $\cos^{-1}(\cos3|x|)$ is equal to

A) $3\pi-9$ B) $10-3\pi$ C) $5\pi-13$ (D) $9-2\pi$

63. If $g(x)=\left(4\cos^4x-2\cos2x-\frac{1}{2}\cos4x-x^7\right)^{\frac{1}{7}}$, then the value of $g(g(100))$ is equal to

A) -1 B) 0 C) 1 D) 100

64. The number of integral values of a for which the equation $\frac{x^2}{a-10}+\frac{y^2}{4-a}=-1$ represents an ellipse are

A) 6 B) 4 C) 5 D) 7

65. The number of ways in which 4 boys and 4 girls can stand in a circle so that each boy and each girl is one after the other is:

A) 3!.4! B) 4!.4! C) 8! D) 7!

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66. If $\sum_{r=2}^{n} \frac{{}^{n-2}C_{r-2}}{{}^{n}C_{r}} = 20$, then n equals :

A) 39 B) 49 C) 59 D) 69

67. Find the equation of the common tangent to the curves $y^2 = 8x$ and $xy = -1$.

A) $y = x + 2$ B) $x = y + 2$ C) $y = 2x + 3$ D) $y = x + 3$

68. If A and B are squares matrices of order 3 such that det. $A = -2$ and det. $B = 1$, then det. $\left(A^{-1}\,\text{adj}\left(B^{-1}\right)\cdot\text{adj}\left(2A^{-1}\right)\right)$ is equal to :

A) 8 B) –8 C) 1 D) –1

69. Two cards are drawn one by one without replacement from a deck of 52 cards. The probability that the second card is higher in rank than the first card, is; (Ranks in increasing order can be taken from Ace to King.)

A) $\frac{1}{17}$ B) $\frac{8}{17}$ C) $\frac{16}{17}$ D) $\frac{9}{17}$

70. If m and n are the smallest positive integers satisfying the relation $\left(2\text{Cis}\frac{\pi}{6}\right)^m = \left(4\text{Cis}\frac{\pi}{4}\right)^n$, then $(m+n)$ has the value equal to : $(cis\theta = cos\theta + isin\theta)$

A) 120 B) 96 C) 72 D) 60

SECTION-II
(NUMERICAL VALUE ANSWER TYPE)

This section contains 5 questions. The answer to each question is a Numerical value. If the Answer in the decimals, **Round off to the nearest Integer only. Have to Answer all 5 questions** and question will be evaluated according to the following marking scheme:
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71. If $ax^3 + bx^2 + cx + d = \begin{vmatrix} x^2 - 2x & x+1 & 3 \\ 2x^2 + x & x-1 & 2 \\ x^2 - 1 & x+2 & 1 \end{vmatrix}$, then a is equal to :

72. If α, β and γ are the roots of the cubic $ax^3 + bx^2 + cx - d = 0$. Point (α, β, γ) lies on the plane $2x + y + z = 7$ and line joining the points (2, –3, 1) and (3, –4, –5). Find value of $\frac{-b + c - d}{a}$.

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73. If $\left(x+\frac{1}{x}\right)(x^2-12|x|-85)\le 0$, Let α be the largest integer & β be the largest negative integer satisfying the inequality then $\alpha+\beta$ is equal to

74. In a class of 30 pupils, 12 take needle work, 16 take physics and 18 take history. If all the 30 students take at least one subject and no one takes all three then the number of pupils taking 2 subjects is α, then $\frac{\alpha}{2}$ is equal to

75. Let α and β are the number of points of discontinuity and non-differentiability of $h(x)$ respectively such that $h(x)=\begin{cases}|x^2+5x-2| & ,-8\le x<0\\ \max(|x|,|x-2|), & 0\le x<2\\ \min(|x-3|,2), & 2\le x<6\\ [2x]\operatorname{sgn}\left(x-\frac{15}{2}\right), & 6\le x\le 8\end{cases}$.

Find the value of $|\alpha-\beta|$. where [.] greatest integer function and sgn(x) denotes signum function.

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